

Muhammad Anas Khan

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Objective:

Computer Science final-year student with hands-on experience in web development using the MERN stack (MongoDB, Express.js, React, Node.js) and building dynamic, component-based UIs with React. Passionate about cross-platform app development and eager to apply growing expertise in Machine Learning and Deep Learning. Seeking a software engineering internship to contribute to innovative, scalable, and intelligent systems in real-world projects.

Education:

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- **Bachelor of Computer Science**, FAST - National University of Computer and Emerging Sciences Khi (2022-Present)
CGPA: 3.04
 - **Pre-Engineering Intermediate**, Govt. Degree Malir Cantt College (2019-2021)
 - **Matriculation** SOS Hermann Gmeiner School (2017-2019)

Experience:

Web Development Intern (MERN Stack) – OutSecure, Rawalpindi

Jul 2025 – Sep 2025

- Completed a structured remote internship program focused on React app development with MERN stack.
- Built beginner-level web projects under guidance to strengthen frontend-backend integration skills.
- Gained exposure to collaborative workflows and industry-standard tools.

Skills:

Programming Languages: Python, C++, C, JavaScript.

Web Development & Frameworks: HTML, CSS, JS, MERN stack, React, FastAPI (Python), Express.js, RESTful API design, TailwindCSS, ShadCN, Redux

Databases & Cloud Services: PostgreSQL, MongoDB, MySQL, OracleSQL, Firebase, AWS

Development Tools: Docker, Git, GitHub, VS Code, Cursor, Ubuntu Linux, JIRA, Android Studio, Google-colab, Microsoft Visual Studio.

Parallel and Distributed programming: OpenMP, MPI.

Machine Learning tools: Numpy, Pandas, Scikit-learn, PyTorch, Keras, Matplotlib.

Speaking Language: English, Urdu

Projects:

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- **Final Year Project – Offline Payment System for Pakistan** (Ongoing, 2025–2026)
 - Developing an **Android (Kotlin)** app enabling **peer-to-peer (P2P)** payments without internet using **QR/BLE-based secure transactions**.
 - Implementing a **Python FastAPI** backend with **double-spending prevention** and **deferred synchronization** logic.

- Designing **cryptographically secure transaction protocols** using asymmetric keys, hash-chains, and Android Keystore hardware protection.

- **Backend Development – Multi-Store Inventory Management System** (2025)
 - Developed a **scalable Express.js backend** with **JWT authentication**, handling 500+ stores, enabling secure access control and role-based permissions for store managers and admins.
 - Designed and optimized a **PostgreSQL database** with centralized product cataloging and storespecific inventory tracking.
 - Implemented **caching (Redis)** and **asynchronous read/write operations** to improve API response times, ensuring high performance during peak transaction loads.
 - GitHub: [Kiryana Store Web Backend](#)
- **Deep Learning Model – Bacterial Colony Classification** (2025)
 - Built an image classifier using EfficientNetB0 with transfer learning, trained on a 33-class bacteria dataset in TensorFlow/Keras (Google Colab), achieving ~14.5% training accuracy
 - Preprocessed dataset by removing corrupt images, applying augmentation, and resizing inputs to 224×224 for improved generalization.
 - Created a Streamlit prototype for image upload and prediction; analyzed overfitting and visualized performance with training/validation curves.
 - GitHub: [Bacteria-Classfier](#)
- **Visual Daily Planner MVP – React + Firebase App** (2025)
 - Developed an MVP daily planner app to support autistic youth in building routines with visual aids, reducing anxiety and improving independence.
 - Built features for creating/reordering tasks with emojis, colors, and progress tracking, with daily resets and history storage in Firestore.
 - Implemented Firebase Authentication with protected routes, Redux state management, and a clean, responsive UI using Shaden.
 - Deployed a mobile-friendly version on Vercel for live testing.
 - GitHub: **Autism-Visual-Routine-Planner** | Vercel: [Live App](#)
- **Frontend Project – Snakey: Classic JavaScript Snake Game** (2025)
 - Built a fully responsive canvas-based Snake game using vanilla JavaScript, HTML5, and CSS to practice frontend fundamentals and dynamic UI manipulation.
 - Implemented real-time keyboard controls, difficulty levels, score tracking, and immersive gameplay by toggling UI visibility during play.
 - Gained hands-on experience with canvas rendering, game loops, and structuring a web project from scratch.
 - GitHub: [Snakey-JS](#)
- **Desktop Game – Pacman (C++ OOP Project)** (2023)
 - Built a classic Pacman game using C++ with object-oriented design and BFS-based ghost AI.
 - Added sound effects, music, difficulty modes, and persistent score tracking via file I/O.
 - Structured with modular classes for game logic, audio, entities, and user interaction.

Interests:

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| • Web development | • Machine Learning/AI |
| • Android Development | • Cyber Security |